Forensic Toxicology (2026) 44:296–312
https://doi.org/10.1007/s11419-025-00754-z

ORIGINAL ARTICLE

Study on the expression changes of hepatic Gluconeogenesis-related proteins induced by insulin Overdose-related liver injury

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Received: 19 August 2025 / Accepted: 13 December 2025 / Published online: 29 December 2025

Abstract

Purpose Insulin overdose severely disrupts hepatic glucose metabolism, leading to toxic changes in the liver that may play an important role in the diagnosis and identification of insulin overdose.

Methods A rat model of insulin overdose was established in this study. ALT and AST levels were measured to assess liver function. Hematoxylin and eosin, oil red O, and periodic acid-Schiff staining were used to evaluate pathological liver injury. Ultrastructural changes in hepatocyte organelles were observed under a transmission electron microscope. The expression levels of liver gluconeogenesis-related proteins, including forkhead box protein O1 (FOXO1) and phosphorylated FOXO1 (p-FOXO1), phosphoenolpyruvate carboxykinase (PEPCK), glucose-6-phosphatase catalytic subunit (G6PC), chromosome region maintenance 1 (CRM1), and tyrosine 3-monooxygenase/tryptophan 5-monooxygenase activation protein (14-3-3), were detected in the rat model and human autopsy liver tissues using immunohistochemistry and immunofluorescence.

Results The insulin overdose group showed elevated ALT/AST, hepatocyte fatty degeneration, glycogen depletion, mitochondrial swelling, and cristae loss. The recovery group exhibited milder fatty changes, glycogen accumulation, and restored mitochondria. Insulin overdose increased hepatic p-FOXO1, CRM1, and 14-3-3 expression but decreased PEPCK and G6PC. FOXO1 shifted from nuclear to cytoplasmic localization in overdose groups, with partial nuclear recovery post-recovery.

Conclusion Insulin overdose leads to hypoglycemia, which in turn causes hepatocyte damage. Insulin overdose promotes the nuclear translocation of FOXO1 protein in hepatocytes, and this process shows high reproducibility in human cases of fatal insulin overdose. The coordinated changes in these indicators have significant value in detecting and diagnosing fatal cases of insulin overdose.

Keywords Insulin overdose · Liver injury · Gluconeogenesis · FOXO1

Introduction

Diabetes is a prevalent disease in modern society. With the development and widespread use of synthetic insulin, it has been widely used globally for its therapeutic effects on diabetes [1]. At physiological concentrations, insulin regulates human metabolism [2]; however, excessive insulin levels can lead to severe hypoglycemia and even death [3, 4]. Additionally, there have been numerous recent reports of cases involving insulin use in suicide and homicides [5, 6].

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Insulin overdose-caused hypoglycemia is a major factor leading to organ damage and even death [7], with the liver being the primary target organ for insulin action and metabolism [8, 9]. Excessive insulin can lead to alterations in hepatic glucose and lipid metabolism, severely inhibiting hepatic gluconeogenesis [10]. As a key target organ for insulin action, the liver is the central site for peripheral glucose uptake and storage and is a hub for lipid metabolism and detoxification functions [11]. Clinically, patients can exhibit elevated transaminases and hepatomegaly following insulin overdose [12–14]. Pathological changes, including hepatocellular edema and necrosis, can be observed in autopsy cases of deaths due to insulin overdose [5, 15].

Gluconeogenesis is a critical process through which insulin regulates hepatic glucose metabolism, leading to hypoglycemia [16]. It is a complex biological process involving various metabolic enzymes. Phosphoenolpyruvate carboxykinase (PEPCK) converts oxaloacetate to phosphoenolpyruvate, while glucose-6-phosphatase catalytic subunit (G6PC) catalyzes the final step of gluconeogenesis; both are key rate-limiting enzymes in this pathway [17]. Gluconeogenesis is regulated by various transcription factors. Insulin activates the phosphatidylinositol 3’ –kinase (PI3K)-Akt signaling pathway, which mediates nuclear translocation of forkhead box protein O1 (FOXO1), thereby inhibiting gluconeogenesis [18]. FOXO1 is an important transcription factor in gluconeogenesis. Under low insulin levels, FOXO1 resides in the nucleus and can activate the expression of gluconeogenic rate-limiting enzymes, including PEPCK and G6PC, promoting glucose production in the liver. When cells respond to insulin signaling, FOXO1 is phosphorylated in the nucleus, forming phosphorylated FOXO1 (p-FOXO1) [19]. Subsequently, with the assistance of chromosomal region maintenance protein 1 (CRM1) [20] and tyrosine 3-monooxygenase/tryptophan 5-monooxygenase activation protein (14-3-3) [21], it is transported to the cytoplasm, losing its activity, which ultimately results in gluconeogenesis inhibition.

This study aimed to establish an insulin overdose rat model, investigate pathological changes in rat livers, assess the gluconeogenesis-related protein expression, and analyze their expression patterns. In addition, we attempted to validate these findings in in autopsy samples from lethal cases of insulin overdose, and provide potential molecular markers with diagnostic value for cases of insulin overdose-related fatalities.

Materials and methods

Experimental animals

Adult healthy male Sprague-Dawley rats, weighing 275–325 g and aged 8–10 weeks, were provided by Hubei Beint Biotechnology Co., Ltd. The rats were randomly divided into three groups: Control, insulin overdose, and insulin overdose recovery (hereafter referred to as the recovery group), with 10 rats in each group. The experimental animals were kept in a quiet environment at 20–25 °C with a humidity of 40%–60%, with regular ventilation, following a 12-h light and 12-h dark cycle. Each cage housed 3–4 rats, and they had free access to food and water during the entire duration. All surgical procedures were conducted following the standards of the Animal Ethics Committee.

Establishment of insulin overdose rat model

After fasting the rats for 16–20 h, their body weights were measured, and blood was collected from the tail tip to assess blood glucose levels. Rats were intraperitoneally injected with 20 IU/kg crystalline human insulin, and tail blood glucose levels were monitored every 30 min while documenting their activity. When the blood glucose dropped below 2.5 mmol/L, the rats were gently placed into an anesthesia induction box for sedation. The initial isoflurane concentration was set at 2.0%, and it was increased to 4.0%–5.0% after approximately 3 min. Subsequently, endotracheal intubation was performed, and electroencephalogram (EEG) monitoring was initiated. Rats in the control group received an equivalent volume of physiological saline intraperitoneally, with all other procedures remaining identical to the experimental group. When rats in the recovery group exhibited signs of hypoglycemia, such as seizures, they were administered 3 mL of 50% glucose solution intraperitoneally in three doses, each administered 5 min apart, while monitoring tail blood glucose levels. After the blood glucose levels rose above 4 mmol/L, the rats were placed back into the cages for normal feeding.

Rat serum samples collection and liver tissue Extraction, Embedding, and sectioning

After EEG monitoring, tissue samples were collected from the experimental and control groups. For the recovery group, samples were collected two days after normal feeding was resumed. A single-use venous blood collection needle was inserted into the right ventricle to collect 4–5 mL of blood using a vacuum blood collection tube. The blood was allowed to stand, and serum was obtained by centrifugation at 4 °C and 2000 rpm for 10 min, after which the supernatant was stored at −80 °C. Small liver tissue samples (1 mm^3) were cut and placed in an electron microscopy fixation solution at 4 °C for further transmission electron microscopy examination. Following perfusion with physiological saline, the liver lobes were separated and stored at −80 °C for later use in frozen sectioning. After perfusion with 4% paraformaldehyde, the liver lobes were placed in tissue fixation solution for 24 h. The tissue was then embedded in embedding molds, dehydrated, cleared, infiltrated with paraffin, and sectioned for subsequent immunohistochemistry and immunofluorescence analyses.

Determination of serum AST and ALT concentrations

After thawing, the serum samples were centrifuged at 4 °C and 3000 rpm for 10 min to obtain the supernatant, which was then mixed with pre-prepared working reagents. The concentrations were automatically measured, and the results were exported using an automatic biochemical analyzer (Rayto, China, Chemray 800). Each sample was measured thrice, and the average value was taken for analysis.

Hematoxylin and Eosin (HE) staining

Liver tissue sections from rats underwent gradient dehydration, followed by immersion in hematoxylin staining solution for 5 min. After washing, the slides were placed in differentiation solution for 5 s and then immersed in bluing solution for 5 s. After washing, the sections were stained with eosin solution for 5 min, dehydrated, cleared, and mounted using neutral gum.

Periodic Acid-Schiff (PAS) staining

Liver tissue sections were subjected to gradient dehydration, and periodic acid solution was added and incubated in the dark for 10 min, followed by washing in distilled water for 5 min. Schiff reagent was then added, and the slides were incubated in the dark at 37 °C for 30 min, followed by another wash in distilled water for 5 min. The sections were then stained with hematoxylin for 30 s, washed, and placed in differentiation solution for 5 s. After washing, the sections were dehydrated, cleared, and mounted with neutral gum.

Oil red O staining

Freshly extracted liver tissues were flattened to make 8 μm frozen sections, fixed in tissue fixation solution for 15 min, and dried. The sections were immersed in oil red O staining solution for 10 min in the dark. They were then differentiated using 60% isopropanol for two rounds of 3 and 5 s, respectively. After two washes with distilled water, the sections were counterstained with hematoxylin for 5 min and washed thrice with distilled water. The differentiation solution was applied for 5 s, followed by two washes with distilled water, and a bluing solution was applied for 1 s before the two final washes. Finally, the sections were mounted using glycerol gel.

Transmission electron microscopy detection of rat hepatocyte mitochondria

Liver tissue blocks fixed in electron microscopy fixative were rinsed thrice with 0.1 M phosphate buffer (pH 7.4), each for 15 min. The samples were then fixed in 1% osmium tetroxide in the dark at room temperature for 2 h and rinsed thrice more. After gradient dehydration with ethanol, the liver tissue was immersed twice in 100% propylene oxide for 15 min each. Following the embedding and sectioning of the liver tissue, toluidine blue staining was performed to locate and adjust the sample under a light microscope. Ultra-thin Sects. (60–80 nm) were cut using an ultra-thin slicer (Leica, Germany, UC7) from the resin block. The sections were picked up on copper grids and stained in the dark with 2% uranyl acetate in saturated alcohol for 8 min; then washed thrice with 70% ethanol and ultrapure water. The sections were stained with 2.6% lead citrate solution for 8 min and washed thrice with ultrapure water. The sections were dried overnight at room temperature. The samples were observed and analyzed under a transmission electron microscope (HITACHI, Japan, HT7800/HT7700).

Immunohistochemistry

The antibodies used for immunohistochemistry are presented in Appendix 1. The sections underwent deparaffinization, hydration, antigen retrieval, blocking of endogenous peroxidase activity, and subsequent blocking. They were then incubated overnight at 4 °C with the primary antibody. The next day, after resetting to room temperature for 15 min, the sections were washed thrice with Tris-Buffered Saline (TBS) buffer and incubated at room temperature with the corresponding secondary antibody for 1 h. Freshly prepared DAB staining solution was used for color development, and hematoxylin was used for counterstaining for 3 min. After dehydration, slides were covered. The slides were scanned using a slide scanner (Hamamatsu, Japan, S360) at the same light intensity. Each slide was randomly photographed in four fields under a 400× high-power microscope, and the mean optical density (MOD) value for each field was calculated using the ImageJ software. The mean of the four values was taken as the MOD value for that slide for subsequent analysis.

Immunofluorescence

The antibodies used for immunofluorescence were the same as those used for immunohistochemistry. After deparaffinization and hydration, the sections were soaked in 0.1% Sudan Black prepared with 70% alcohol for 30 min and then washed thrice. After blocking, the sections were incubated overnight at 4 ℃ with the primary antibody. The next day, after resetting to room temperature for 15 min, the sections were washed thrice with TBS buffer and incubated at room temperature with the corresponding secondary antibody for 90 min, avoiding light throughout the process. After washing, DAPI staining solution was added, and the sections were stained at room temperature for 10 min, followed by washing. An anti-fluorescence quenching mounting agent was added to complete mounting. Each slide was photographed in six randomly selected areas at high magnification (400×), and the images were imported into the ImageJ software to calculate the MOD value for each photo. The average of the MOD values of all six photos was considered the MOD value for that slide for subsequent analysis.

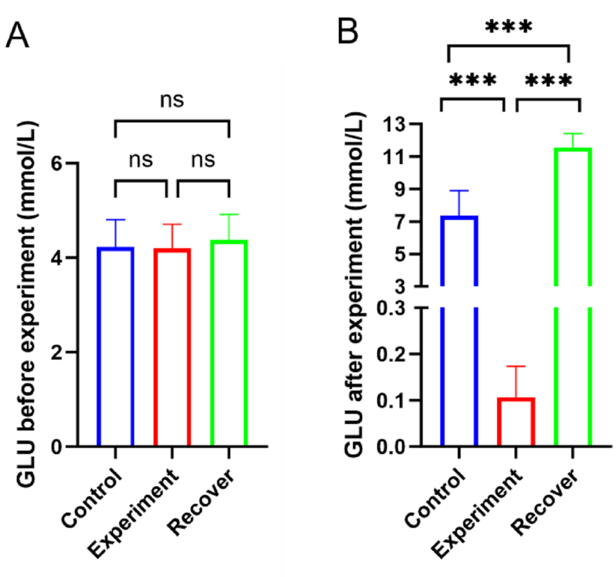

Fig. 1 Changes in blood glucose levels in rats measured before and after the experiment. (**A**) Before the experiment, the blood glucose levels of the three groups of rats were generally consistent. (**B**) After the insulin injection, the blood glucose levels in the insulin overdose group decreased significantly. Treatment with the glucose solution resulted in a significant increase in blood glucose levels in the recovery group. ns: $P>0.05$, ***: $P<0.001$

Autopsy sample selection

Autopsy tissue samples were obtained from the Chinese Human Tissue Bank (Department of Forensic Medicine, Huazhong University of Science and Technology, Wuhan, China). Three cases of insulin overdose death and three control cases matched for age, sex, and cause of death were selected. The case information is provided in Appendix 2. All human tissues were derived from body donors, ensuring no risk to the living individuals. To maintain donor privacy, each specimen was assigned a de-identified numeric code, and all personally identifiable information (such as names and ID numbers) was permanently anonymized. Written informed consent was obtained from the legal representatives of the donors, and the research protocol was approved by the Ethics Committee of the Huazhong University of Science and Technology.

Statistical analysis

Data analysis and chart preparation were performed using GraphPad Prism software (version 9.0.0). Statistical analysis was conducted using an independent samples t-test, with $P<0.05$ considered statistically significant. Statistical results are expressed as the mean ± standard error of the mean.

Results

General condition of rats with insulin overdose

Approximately 1.5–2.0 h after the intraperitoneal injection of insulin, the blood glucose levels in rats gradually decreased to 2.0 mmol/L or below. This was accompanied by symptoms, including limb weakness, significantly reduced activity. As blood glucose levels continued to decline, the rats exhibited signs of urinary and fecal incontinence, convulsions, righting reflex loss, and generalized tonic-clonic seizures with opisthotonos. The changes in the blood glucose levels of the rats before and after the experiment are presented in Fig. 1. There were statistically non-significant differences in blood glucose levels among the three groups of rats before the experiment. After insulin injection, the blood glucose levels in the insulin overdose group significantly decreased, and the blood glucose levels in the control group exhibited a stress-induced increase due to the sham operation. Conversely, the combination of stress factors and high-concentration glucose solution led to a marked increase in blood glucose levels in the recovery group.

Changes in serum ALT and AST levels in rats with insulin overdose

ALT and AST serum levels in rats are presented in Fig. 2. Compared to the control group, the insulin overdose group exhibited elevated serum levels of ALT and AST, indicating liver tissue damage in the insulin overdose group. Conversely, the serum ALT and AST levels in the recovery group were lower than those in the insulin overdose group, with non-significant differences between the recovery and control groups.

Pathological changes in liver tissue of rats with insulin overdose

HE, oil red O, and PAS staining were performed to assess the pathological changes in the liver of the three groups of rats, and the results are demonstrated in Fig. 3. HE staining results indicated that liver cells in the control group were closely and neatly arranged without obvious vacuolation, degeneration, or necrosis, and the tissue structure appeared normal. In the insulin overdose group, the liver exhibited fatty degeneration as the main pathological feature, with numerous round, clear vacuoles (fat droplets) of varying sizes visible in liver cell cytoplasm. Most liver cell nuclei remained centrally located, demonstrating microvesicular fatty degeneration. Fatty degeneration was predominantly observed in the liver cells surrounding the central vein and was distributed diffusely. Some liver cells were swollen and exhibited focal necrosis. In the recovery group, diffuse small vacuoles were observed within the liver cells. However, the degree of fatty degeneration was lower than that in the insulin overdose group. The oil red O staining results demonstrated no signs of fatty degeneration in the liver of the control group. However, the liver of the insulin overdose group demonstrated strong positive staining, indicating diffuse fatty degeneration of liver cells. In the recovery group, a small number of liver cells exhibited fatty degeneration. PAS staining results demonstrated that non-significant glycogen accumulation was observed in the liver cells of control and insulin overdose groups. Conversely, substantial glycogen accumulation was observed in the liver cells of the recovery group.

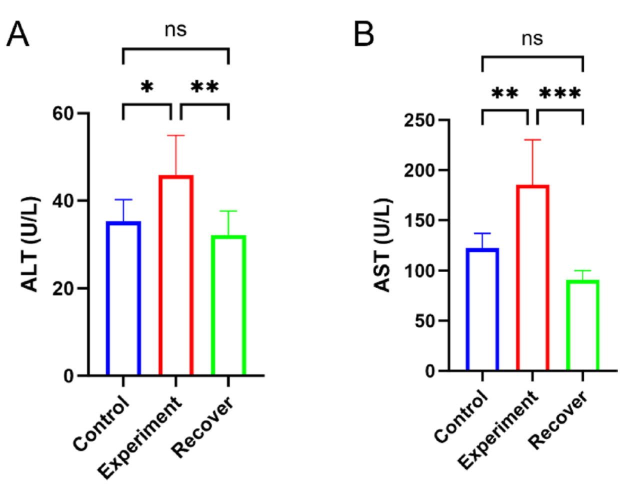

Fig. 2 Serum expression levels of ALT (Fig. 2**A**) and AST (Fig. 2**B**) in rats. The serum ALT and AST levels of rats in the insulin overdose group increased, while the levels in the recovery group slightly decreased. *: $P<0.05$; **: $P<0.01$; *: $P<0.001$

Ultrastructural changes in rat hepatocytes with insulin overdose

The results of the electron microscopy observations of hepatocytes from the three groups of rats are presented in Fig. 4. In the control group, the hepatocyte membranes were intact, with abundant cytoplasmic matrix and a small number of round lipid droplets observed, along with slight swelling of some organelles. The nuclei were oval-shaped with intact nuclear membranes, and the nucleoli were closely associated with the nuclear membrane, exhibiting peripheral aggregation changes, while the chromatin was evenly distributed. The mitochondria exhibited mild swelling and degeneration, with no membrane rupture observed, although some mild dissolution of the cristae structure was observed, with most remaining intact and presenting focal dissolution phenomena. The rough endoplasmic reticulum (RER) structure appeared normal, depicting a parallel bilayer membrane arrangement with ribosomes attached to the surface, and there were no signs of rupture, dilation, or granule loss. A few primary lysosomes and autophagosomes were visible, and the canalicular structures around hepatocytes were normal.

In the insulin overdose group, hepatocyte membranes were blurred, and many lipid droplet structures were visible within the cytoplasm. The organelles exhibited significant swelling and vacuolation. The nuclei were round with intact nuclear membranes, but nucleoli were not observed. The mitochondria exhibited moderate swelling and degeneration, with partial rupture of the outer membrane. The cristae structures were mostly dissolved, with a decrease in the electron density of the mitochondrial matrix. The RER structure was normal, exhibiting a parallel bilayer membrane arrangement with ribosomes throughout the surface without signs of rupture, dilation, or granule loss. Many primary lysosomes and a small number of secondary lysosomes were observed.

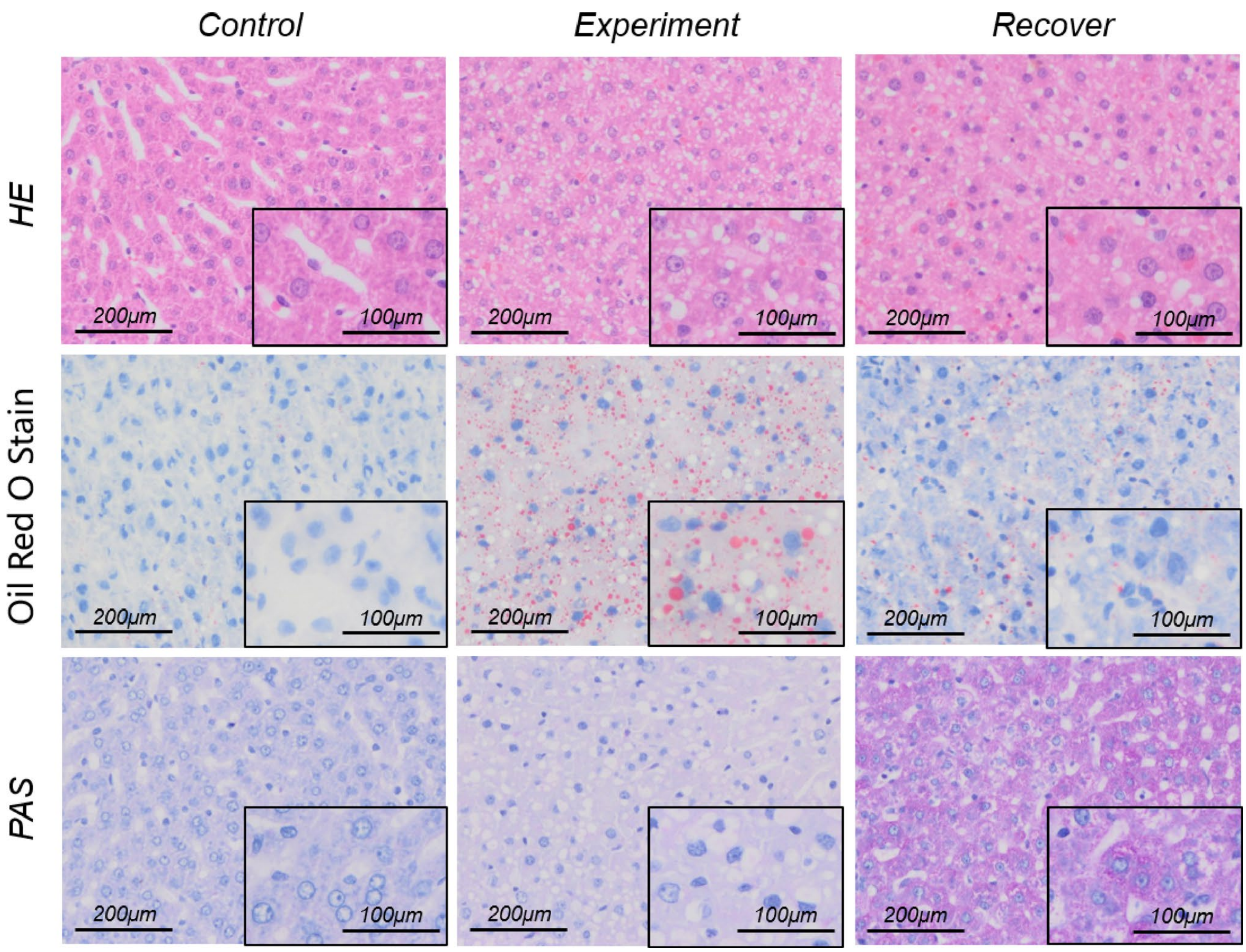

Fig. 3 Pathological changes in rat liver tissue. The liver cell tissue structure in the control group was generally normal, while the insulin overdose group showed fatty degeneration and fat vacuoles in liver cells. In the recovery group, some liver cells exhibited fatty degeneration, accompanied by significant glycogen accumulation

In the recovery group, hepatocyte membranes were intact, with many glycogen pools visible within the cytoplasm. The nuclei were round with intact membranes, centrally located nucleoli, and evenly distributed chromatin. The mitochondrial structure appeared largely normal without rupture of the inner or outer membranes. The cristae structures were clear without dissolution observed, while the electron density of the mitochondrial matrix was normal. The RER structure was normal, exhibiting a parallel bilayer membrane arrangement with ribosomes attached to the surface without signs of rupture, dilation, or granule loss. The canalicular structures around the hepatocytes were normal.

Changes in FOXO1/p-FOXO1 expression in liver tissue of rats with insulin overdose

The immunohistochemical (IHC) results for FOXO1 and p-FOXO1 ($n=3$ per group) are presented in Fig. 5. In the control, insulin overdose, and recovery groups, FOXO1 was positively expressed in hepatocytes. In the control group, the positive FOXO1 signals were primarily concentrated in hepatocyte nuclei. Conversely, in the insulin overdose group, the positive FOXO1 signals were predominantly observed in the cytoplasm, with a non-significant difference in FOXO1 protein expression levels between the two groups (FOXO1: 0.9839-fold experiment vs. control, $p=0.9168$). The hepatocytes in the recovery group displayed strong positive expression of FOXO1 in the nuclei and cytoplasm (FOXO1: 1.389-fold recover vs. control, $p=0.0533$). The positive p-FOXO1 expression level in the hepatocytes of the insulin overdose group was higher than that in control and recovery groups (p-FOXO1: 2.098-fold experiment vs. control, $p=0.0038$; 1.244-fold recover vs. control, $p=0.0995$).

The immunofluorescence results for FOXO1 ($n=3$ per group) are demonstrated in Fig. 6. The immunofluorescence staining indicated positive FOXO1 expression in the

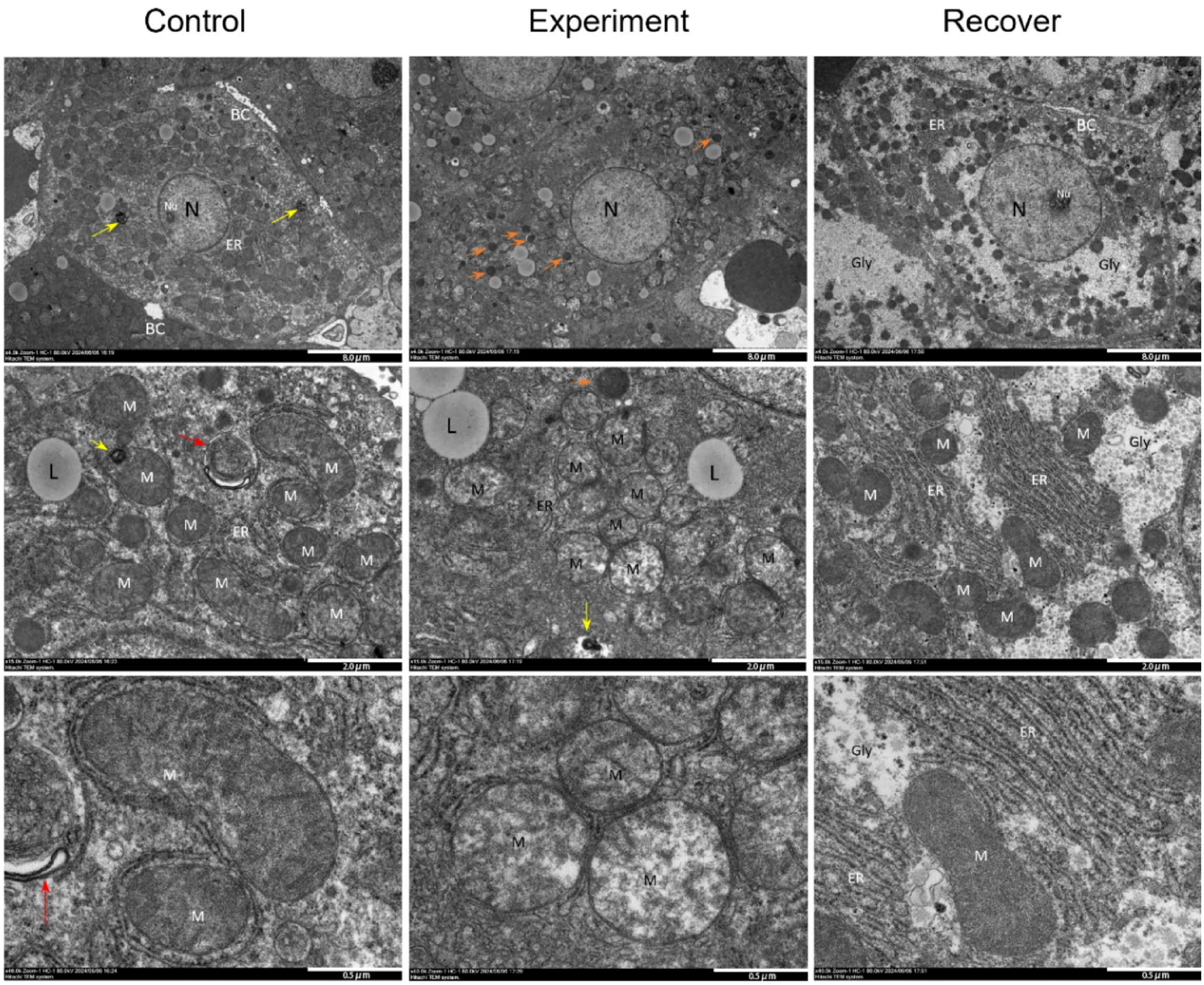

Fig. 4 Transmission electron microscopy results of rat hepatocytes. N, nucleus; Nu, nucleolus; L, lipid droplet; M, mitochondria; ER, endoplasmic reticulum; BC, bile canaliculus; Gly, glycogen; orange arrow, primary lysosome; yellow arrow, secondary lysosome; red arrow, mitochondrial lysosome

hepatocytes of the control, insulin overdose, and recovery groups. In the control group, FOXO1 fluorescence was primarily concentrated in the hepatocyte nuclei, with weaker expression in the cytoplasm. In the insulin overdose group, FOXO1 fluorescence was predominantly observed in the cytoplasm, with weaker expression in the nuclei, and there was a non-significant difference in protein expression levels between the two groups (FOXO1: 0.9829-fold experiment vs. control, p=0.8171). In the recovery group, FOXO1 was positively expressed in the nuclei and cytoplasm of hepatocytes, with an increased protein expression level compared to that in the control and insulin overdose groups (FOXO1:1.261-fold recover vs. control, p=0.0171). The immunofluorescence results were consistent with immunohistochemistry findings.

Changes in PEPCK and G6PC expression in liver tissue of rats with insulin overdose

The results of the expression changes for PEPCK and G6PC (n=3 per group) are exhibited in Fig. 7. IHC results demonstrated a strong positive expression of PEPCK and G6PC in the hepatocytes of the control and recovery groups (PEPCK: 0.9585-fold recover vs. control, p=0.3424; G6PC: 1.093-fold recover vs. control, p=0.0318). However, PEPCK and G6PC expressions were decreased in the hepatocytes of the insulin overdose group (PEPCK: 0.7098-fold experiment vs. control, p<0.0001; G6PC: 0.6149-fold experiment vs. control, p=0.0002).

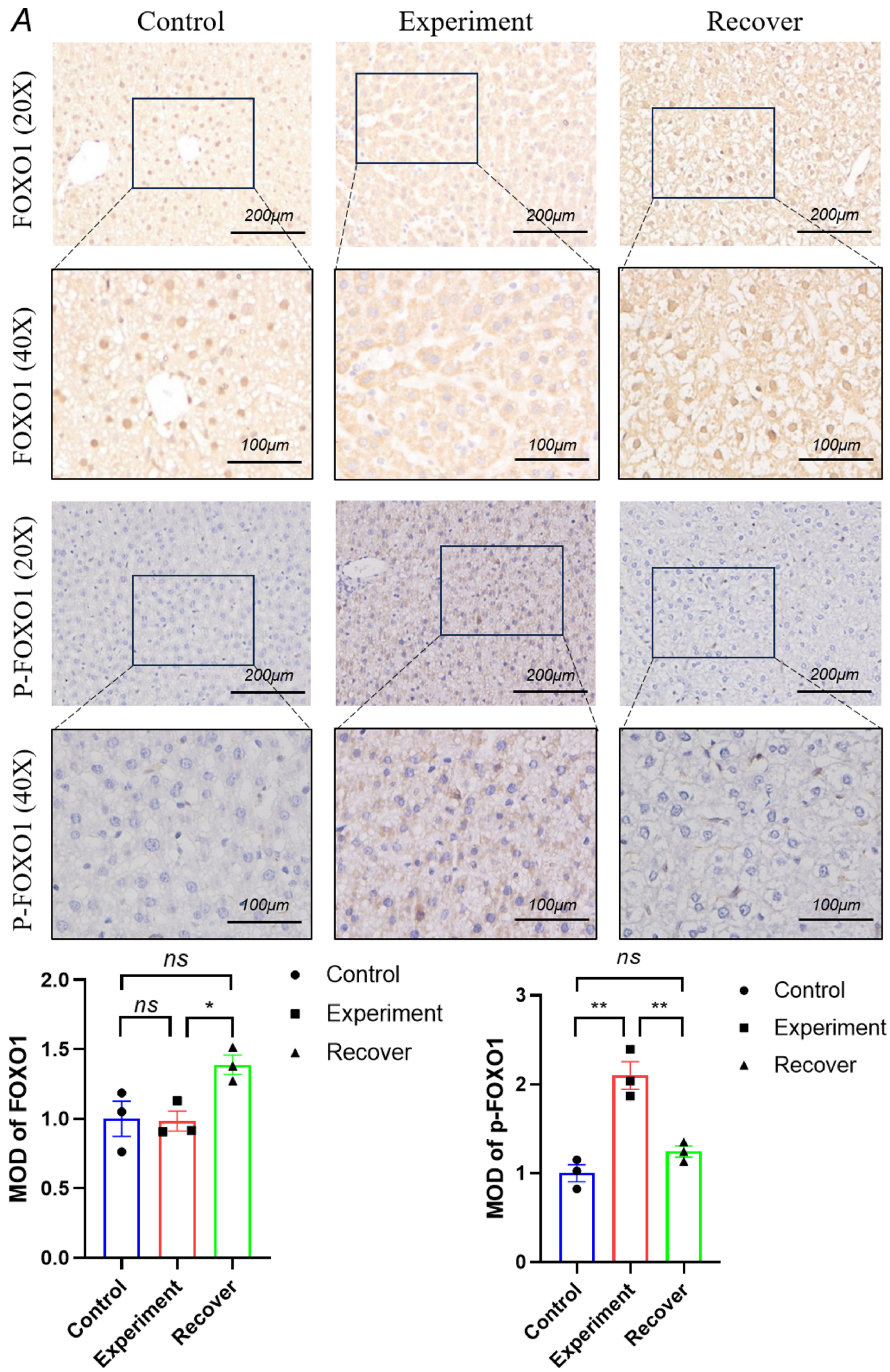

Fig. 5 IHC staining results of FOXO1 and p-FOXO1 in rat liver. In the control group, positive FOXO1 signals were concentrated in hepatocyte nuclei, while the signals in insulin overdose group were localized in the cytoplasm. In the recovery group, FOXO1 was observed in both the nuclei and the cytoplasm of liver cells, with expression levels higher than those in the control and insulin overdose groups. The positive expression levels of p-FOXO1 in the hepatocytes of the insulin overdose group were higher than those in the control and recovery groups. *: $P<0.05$; **: $P<0.01$

Changes in CRM1 and 14-3-3 expression in liver tissue of rats with insulin overdose

The results of the expression changes for CRM1 and 14-3-3 (n=3 per group) are presented in Fig. 8. IHC results revealed that, compared to the control group, CRM1 and 14-3-3 expressions were enhanced in the hepatocytes of the insulin overdose group (CRM1: 2.058-fold experiment vs. control, p=0.0016; 14-3-3:1.824-fold experiment vs. control, p=0.0111). Concurrently, there were non-significant changes in CRM1 and 14-3-3 expression levels in the recovery group (CRM1: 0.8961-fold recover vs. control, p=0.3813; 14-3-3: 0.7547-fold recover vs. control, p=0.1401).

IHC Detection of FOXO1/p-FOXO1, PEPCK, G6PC, CRM1, and 14-3-3 in Liver Tissue Samples from Insulin Overdose Fatal Cases

IHC staining was performed to validate the expression of FOXO1, p-FOXO1, PEPCK, G6PC, CRM1, and 14-3-3 in liver samples from three cases of fatal insulin overdose and three control cases. The results are presented in Fig. 9. Positive expression of FOXO1 was observed in the liver samples from the control group and fatal insulin overdose cases (FOXO1: 0.9315-fold insulin overdose vs. control, p=0.4784). FOXO1 staining in liver samples from the fatal insulin overdose cases were primarily concentrated in the cytoplasm, with less evident staining in the nuclei. p-FOXO1 levels were significantly elevated in the insulin overdose group (p-FOXO1: 2.79-fold insulin overdose vs. control, p<0.0001). PEPCK and G6PC expression levels were reduced in the liver samples from the fatal insulin overdose cases (PEPCK: 0.5198-fold insulin overdose vs. control, p=0.0026; G6PC: 0.3349-fold insulin overdose vs. control, p<0.0001), while CRM1 and 14-3-3 expression levels were elevated (CRM1: 2.329-fold insulin overdose vs. control, p=0.0033; 14-3-3: 1.869-fold insulin overdose vs. control, p=0.0011). IHC results from the human liver samples were consistent with the findings of animal experiments.

Discussion

In lethal insulin overdose cases, the liver tissue can exhibit severe edema and mild-to-moderate necrosis [22]. Serum ALT and AST levels are commonly used to assess the extent of liver damage [23]. Our results demonstrate that, compared to the control and recovery groups, ALT and AST levels in the insulin overdose group of rats were significantly elevated, demonstrating that insulin overdose causes hepatocellular injury in rats. However, high-concentration glucose injection can alleviate insulin-induced liver damage to some extent. HE staining of the liver revealed that the damage from insulin overdose primarily manifests as microvesicular fatty degeneration, with numerous vacuolated lipid droplets present in liver cells, and the nuclei of hepatocytes remaining centrally located. Timely injection of high-concentration glucose can reduce lipid accumulation in the liver tissue and alleviate hepatocellular fatty degeneration, as further validated by oil red O staining. Transmission electron microscopy revealed structural damage in the mitochondria of hepatocytes under hypoglycemic conditions, characterized mainly by mitochondrial swelling, partial rupture of the outer membrane, and numerous fractured cristae, with most of the matrix dissolved.

Hepatic gluconeogenesis is a complex physiological process that maintains blood-glucose homeostasis. FOXO1 is a key transcription factor that activates gluconeogenesis by influencing the transcription of key genes and regulating multiple biological processes [24]. Under hypoglycemic conditions, FOXO1 operates within the nucleus; however, when blood glucose levels rise and insulin is secreted, phosphorylated AKT promotes the FOXO1 translocation from the nucleus to the cytoplasm through the coordinated action of CRM1 and 14-3-3 proteins, thereby losing its regulatory capacity over gluconeogenesis [25]. Our research findings indicate the differences in cellular localization of FOXO1. In control group rats, FOXO1 is mainly distributed in the nucleus of hepatocytes, whereas in the insulin overdose group, FOXO1 is predominantly localized in the cytoplasm of hepatocytes. In the recovery group, positive FOXO1 staining can be observed in the nucleus and cytoplasm of hepatocytes. The p-FOXO1 expression level in the liver tissue of rats in the insulin overdose group is significantly elevated, with a non-significant difference between recovery and control groups. This finding suggests that under conditions of insulin overdose, FOXO1 undergoes phosphorylation in response to insulin signaling, leading to its translocation from the nucleus to the cytoplasm, resulting in a loss of regulatory control over PEPCK and G6PC gene transcription. After a timely and effective rescue with the injection of a high-concentration glucose solution, the insulin level in the rats decreased due to metabolic degradation, resulting in FOXO1 dephosphorylation and its return to the nucleus, restoring its regulatory function in gluconeogenesis.

As key rate-limiting enzymes in the hepatic gluconeogenesis pathway, PEPCK and G6PC can reflect the level of hepatic gluconeogenesis to some extent [26]. Insulin mediates AKT phosphorylation to inhibit PEPCK gene expression, thereby reducing glucose production [27]. G6PC, a membrane-bound protein mainly located in the endoplasmic reticulum [28], specifically catalyzes the hydrolysis of glucose-6-phosphate to produce glucose and phosphate and serves as the key enzyme in the last critical gluconeogenesis reaction []. G6PC activity is also regulated by various factors, with hormonal regulatory mechanisms similar to those of PEPCK [29]. Our results

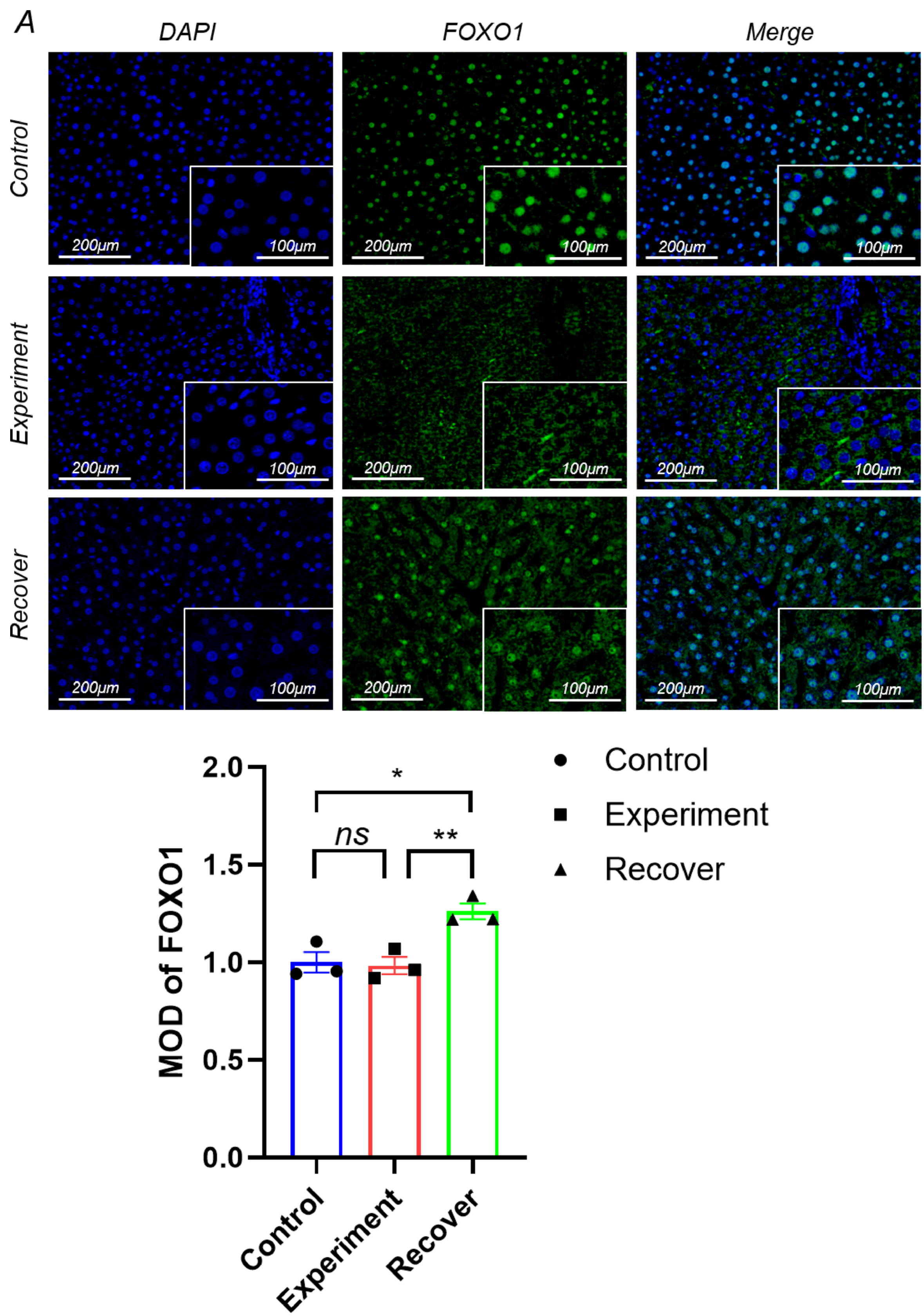
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Fig. 6 Immunofluorescence staining results of FOXO1 in rat liver. In the control group, FOXO1 fluorescence was primarily concentrated in the nuclei of hepatocytes. In the insulin overdose group, FOXO1 fluorescence was predominantly localized in the cytoplasm, with no significant difference in expression levels between the two groups. In the recovery group, FOXO1 fluorescence exhibited positive expression in both the nuclei and the cytoplasm of hepatocytes, with overall expression levels higher than those in the control and insulin overdose groups

demonstrate that, compared to the control group, PEPCK and G6PC protein expression levels in the insulin overdose group of rats exhibited a declining trend. This further illustrates that insulin overdose inhibits rate-limiting enzyme expression in gluconeogenesis, reducing gluconeogenesis rate and ultimately leading to severe hypoglycemia.

FOXO1 nuclear translocation is a complex process, with CRM1 and 14-3-3 being key molecules among them [21]. CRM1, belongs to a family of nuclear transport receptors, with a C-terminal region rich in leucine serving as an NES-binding site [30, 31]. The FOXO1 protein contains an NES; when the cell is stimulated by specific signals, CRM1 binds to Ran-GTP to form a complex that recognizes and binds the NES of FOXO1, transports FOXO1 from the nucleus to the cytoplasm [32]. The 14-3-3 protein is a highly conserved acidic protein that usually exists in dimeric form [33]. When FOXO1 is phosphorylated, its phosphorylation sites can bind to the 14-3-3 protein, which masks the nuclear localization signal of FOXO1 and inhibits its ability to enter the nucleus [34]. Conversely, this binding can cooperate with CRM1 or directly promote the interaction between FOXO1 and CRM1, accelerating the transfer of FOXO1 from the nucleus to the cytoplasm [35]. Our results indicate that, compared to the control group, there is an upward trend in the protein expression levels of CRM1 and 14-3-3 in the liver tissues of rats in the insulin overdose group. This finding suggests that after insulin overdose, with the occurrence of FOXO1 nuclear translocation, CRM1 and 14-3-3 expression increase, jointly participating in hepatic gluconeogenesis inhibition and ultimately leading to hypoglycemia. After the injection of a high-concentration glucose solution into the rats, FOXO1 in the liver tissues underwent dephosphorylation and was re-transported to the nucleus, leading to decreased CRM1 and 14-3-3 expression, allowing for the activation of gluconeogenesis-related genes and ultimately raising blood glucose levels.

In autopsy liver samples, FOXO1 in the control group was mainly localized in the hepatocyte nuclei. In contrast, in the liver samples from insulin overdose cases, FOXO1 was predominantly distributed in the cytoplasm, with a non-significant positive staining observed in the nucleus, and an increase in p-FOXO1 protein expression. Meanwhile, PEPCK and G6PC expression decreased, while the CRM1 and 14-3-3 expression increased, which is consistent with the animal experiments. This study systematically verified the spatiotemporal dynamics of FOXO1 nuclear-cytoplasmic translocation and inhibition of gluconeogenesis in human autopsy samples from fatal insulin overdose cases, thereby connecting classical metabolic mechanisms with forensic toxicology practice. Although the insulin FOXO1 signaling pathway has been well documented, this research integrated animal models and human autopsy samples to reveal specific manifestations of this pathway in post-mortem tissues: the phosphorylation status of FOXO1, CRM1/14-3-3-mediated nuclear-cytoplasmic transport, and the downstream inhibition of gluconeogenic enzymes all showed high reproducibility in fatal insulin overdose cases. This finding provides molecular-level experimental evidence for forensic identification, overcoming the limitations of previous studies that primarily focused on physiological or chronic disease models.

However, our study has certain limitations. First, the small sample size of human autopsy cases (n=3) may not fully account for individual differences in insulin overdose cases, such as the duration of diabetes, concurrent liver diseases, and other factors. Second, the impact of post-mortem interval and terminal metabolic status on the stability of phosphorylated proteins has yet to be quantified. Although animal models indicate that molecular changes persist, the impact of protein degradation on detection results in human samples still needs to be assessed through additional control experiments.

In summary, this study explored the potential value of FOXO1 subcellular localization and phosphorylation status as supplementary diagnostic markers for insulin overdose through comparative analysis of animal models and human cases. In the future, with a larger sample size and the establishment of quantitative detection standards, rapid identification protocols based on liver tissue sections may be developed. Such protocols would be particularly useful for insulin overdose cases where blood insulin concentration testing is limited or where there is a complex medication history, thereby providing a more comprehensive molecular evidence chain for forensic pathology.

Conclusion

Insulin overdose leads to hypoglycemia, which in turn causes hepatocyte damage, primarily manifested as fatty degeneration and mitochondrial injury. Insulin overdose promotes the nuclear translocation of FOXO1 protein in hepatocytes, enhancing its transcriptional activation of key gluconeogenic enzymes, including PEPCK and G6PC, thereby affecting gluconeogenesis. This nuclear translocation process shows high reproducibility in human cases of fatal insulin overdose. The nucleocytoplasmic transport of FOXO1/p-FOXO1 and the coordinated changes in indicators such as PEPCK, G6PC, CRM1, and 14-3-3 have significant value in detecting and diagnosing fatal cases of insulin overdose.

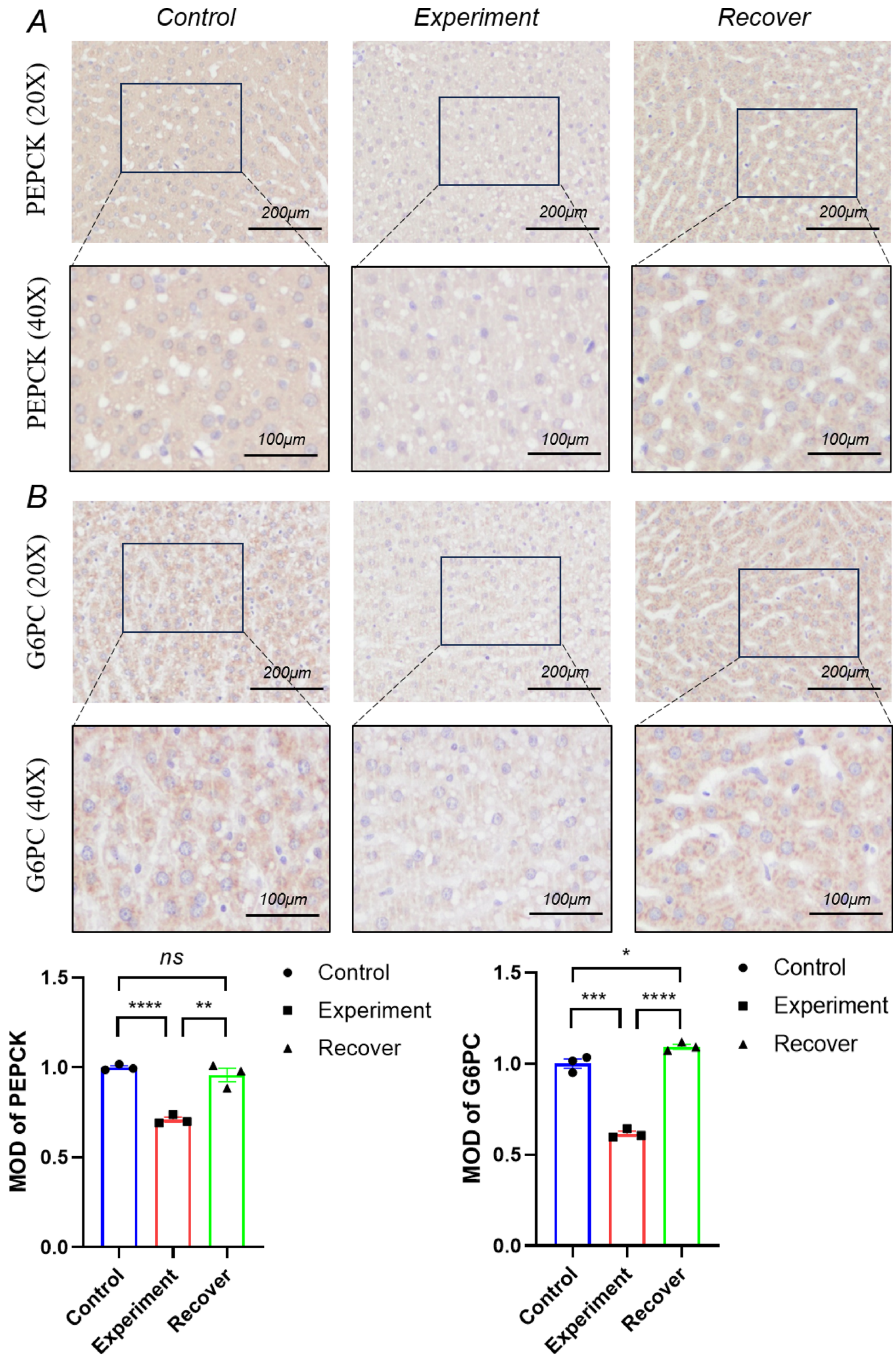

Fig. 7 Expression changes of PEPCK (Fig. 7**A**) and G6PC (Fig. 7**B**) in rat liver (20 and 40×). Compared to the control group, the expression of PEPCK and G6PC in hepatocytes of the insulin overdose group was reduced. There was no significant difference between the recovery group and the control group. *: $P<0.05$; **: $P<0.01$; ***: $P<0.001$

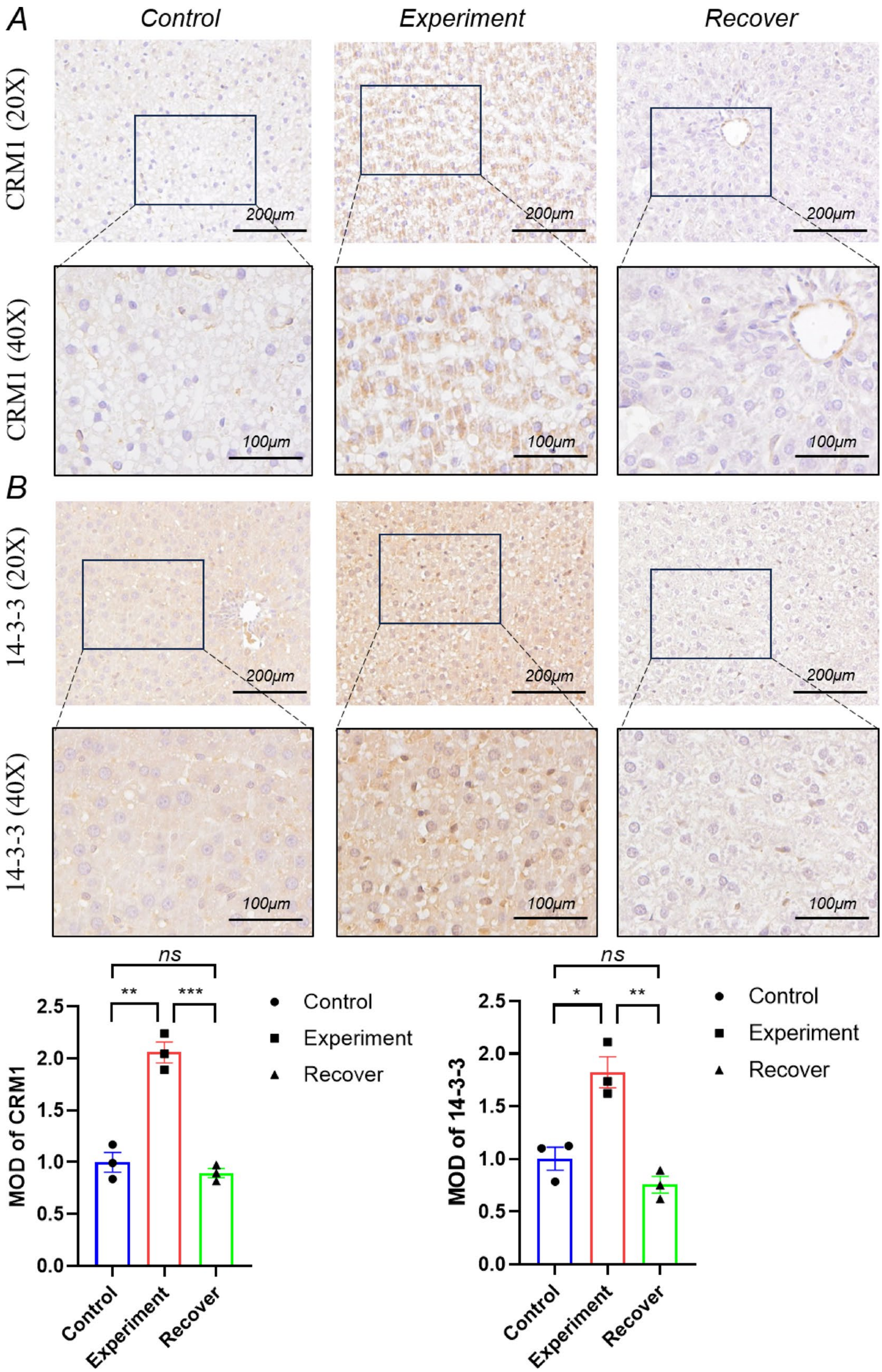

Fig. 8 Expression changes of CRM1 (Fig. 8A) and 14-3-3 (Fig. 8B) in rat liver (20 and 40×). Compared to the control group, CRM1 and 14-3-3 expressions were enhanced in the hepatocytes of the insulin overdose group. There were non-significant changes in CRM1 and 14-3-3 expression levels in the recovery group. *: $P<0.05$; **: $P<0.01$; ***: $P<0.001$

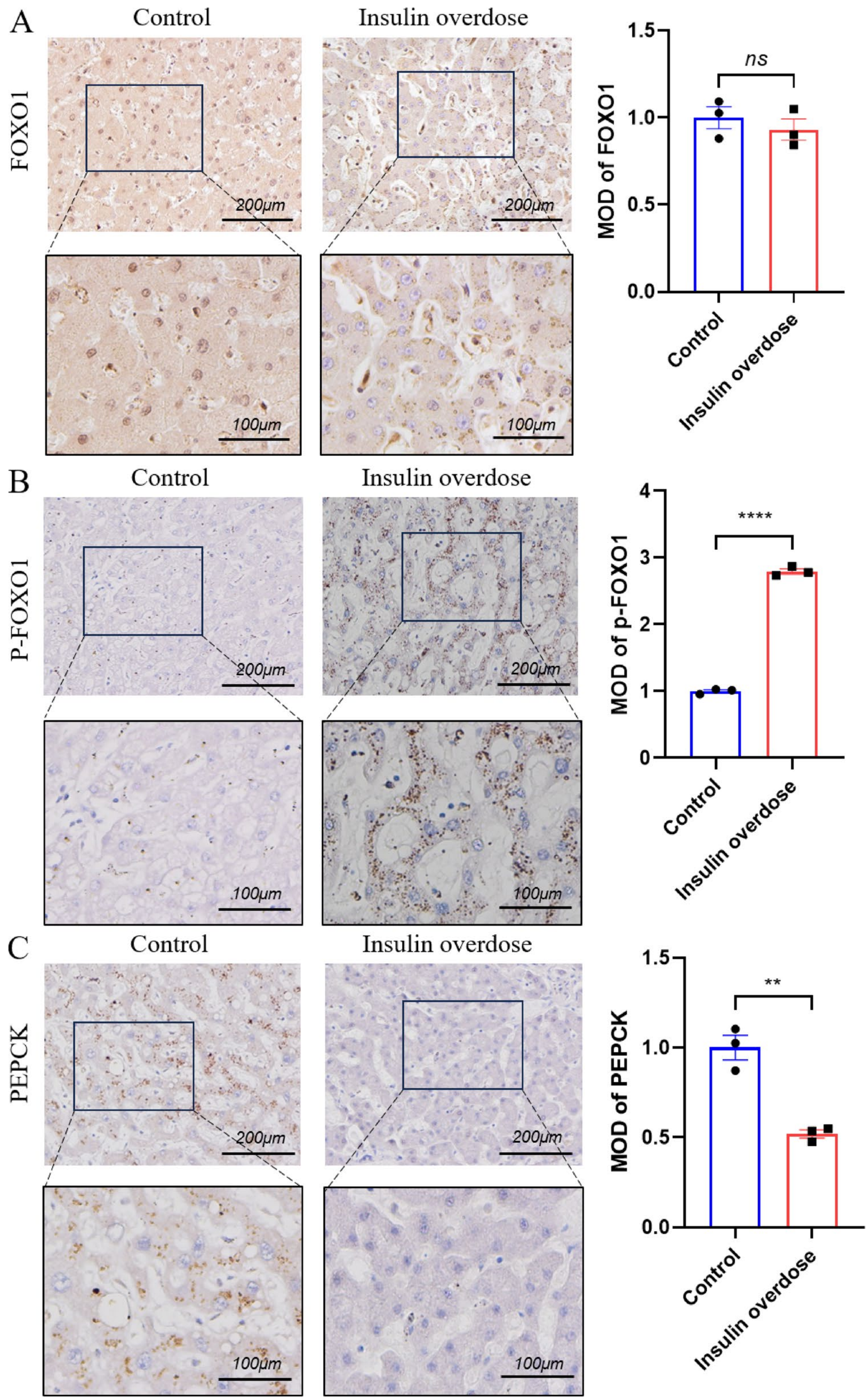

Fig. 9 Expression of FOXO1 (Fig. 9A), p-FOXO1 (Fig. 9B), PEPCK (Fig. 9C), G6PC (Fig. 9D), CRM1 (Fig. 9E), and 14-3-3 (Fig. 9F) in human liver samples (20 and 40×). Compared to the control group, there was no significant difference in FOXO1 expression in human hepatocytes of the insulin overdose group, while the expression of p-FOXO1, CRM1, and 14-3-3 was increased, and the expression of PEPCK and G6PC was reduced.*: $P<0.05$; **: $P<0.01$; ***: $P<0.001$

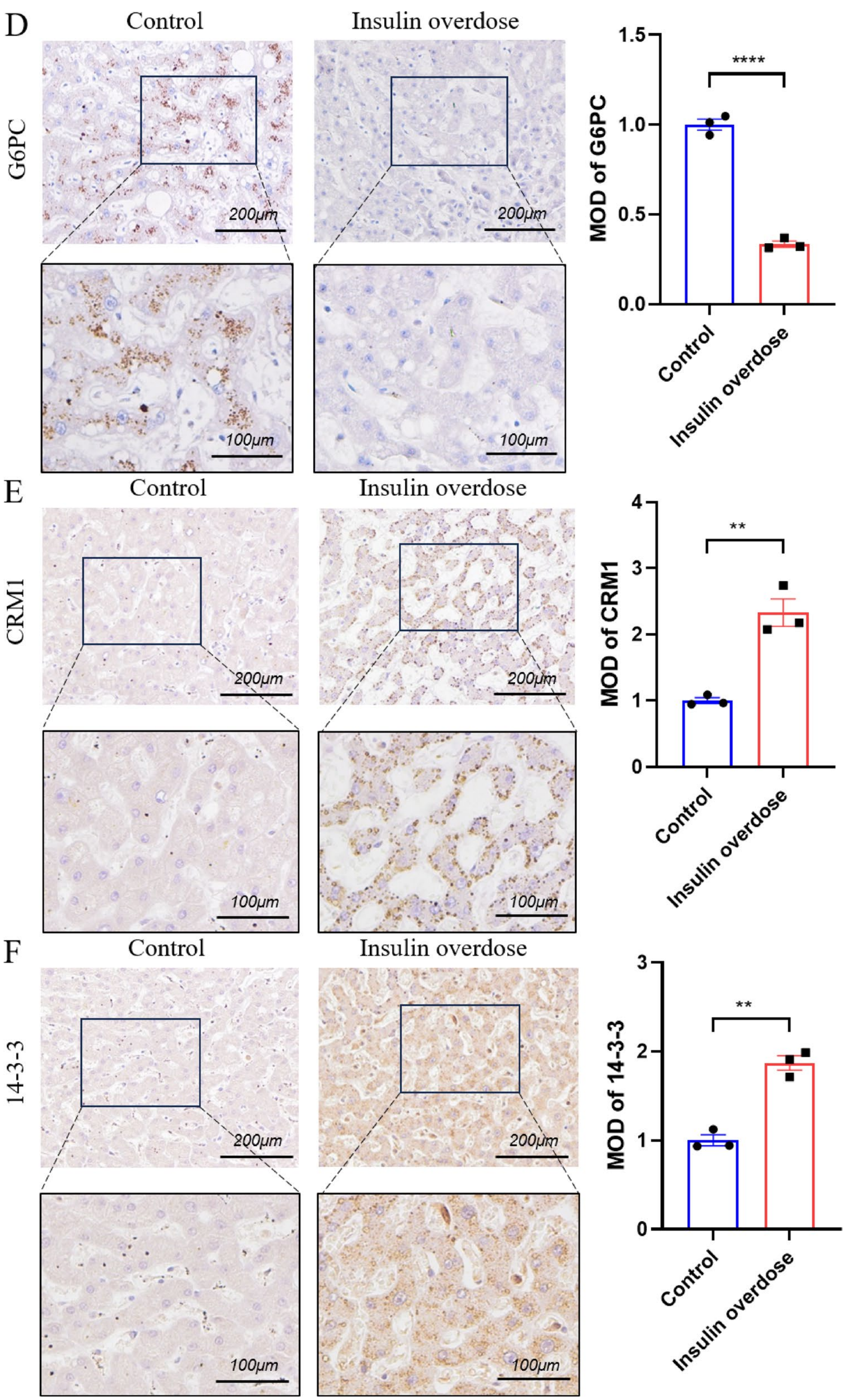

Fig. 9 (continued)

Appendix 1

Antibodies used in immunohistochemistry and immunofluorescence.

Target	Species	Dilution ratio	Supplier	Article No.
FOXO1	rabbit	1:200	Boster, China	BM4249
p-FOXO1	rabbit	1:200	Affinity, China	AF3419
G6PC	rabbit	1:250	Proteintech, England	25771-1-AP
PEPCK	rabbit	1:200	Boster, China	A02022-3
CRM1	rabbit	1:400	Abclonal, China	A19625
14-3-3	mouse	1:500	Proteintech, England	67698-1-lg

Appendix 2

Autopsy sample information.

Group	Cause of death	Age	Gender	Post-mortem interval
Experiment	Insulin overdose	39	Female	4 d
Experiment	Insulin overdose	22	Male	7 d
Experiment	Insulin overdose	36	Female	24 h
Control	Coronary heart disease	35	Male	2 d
Control	Coronary heart disease	44	Female	4 d
Control	Brain injury	38	Female	24 h

Declarations

Competing interests The authors declare that they have no conflicts of interest.

Ethics approval This study involving human participants and/or animals was reviewed and approved by Ethics Committee of the Department of Forensic Medicine, Huazhong University of Science and Technology.

References

1. Sims EK, Carr A, Oram RA, DiMeglio LA, Evans-Molina C (2021) 100 years of insulin: celebrating the past, present and future of diabetes therapy. Nat Med 27(7):1154–1164. https://doi.org/10.1038/s41591-021-01418-2
2. Norton L, Shannon C, Gastaldelli A, DeFronzo RA (2022) Insulin: the master regulator of glucose metabolism. Metabolism 129:155142. https://doi.org/10.1016/j.metabol.2022.155142
3. Geller AI, Shehab N, Lovegrove MC, Kegler SR, Weidenbach KN, Ryan GJ, Budnitz DS (2014) National estimates of insulin-related hypoglycemia and errors leading to emergency department visits and hospitalizations. JAMA Intern Med 174(5):678–686. https://doi.org/10.1001/jamainternmed.2014.136
4. Bottinelli C, Cartiser N, Bevalot F, Fanton L, Guitton J (2020) Is insulin intoxication still the perfect crime? Analysis and interpretation of postmortem insulin: review and perspectives in forensic toxicology. Crit Rev Toxicol 50(4):324–347. https://doi.org/10.1080/10408444.2020.1762540
5. Yuan Y, Yu Z, Tong F, Zhao S, Li Y, Shi Q, Zhou Y (2024) A retrospective study of 29 fatal cases of insulin overdose. Forensic Sci Int 361:112126. https://doi.org/10.1016/j.forsciint.2024.112126
6. Li Y, Yu Z, Yuan Y, Zhang J, Zhao S, Duan Y, Zhou Y (2024) Two cases of fatal insulin homicide resolved using combined immunohistochemistry and immunofluorescence techniques. Forensic Sci Med Pathol. https://doi.org/10.1007/s12024-024-00921-5
7. Yuan Y, Liu Y, Wang S, Zhang J, Gao X, Li Y, Yu Z, Zhou Y (2025) Tryptophan-derived metabolites and glutamate dynamics in fatal insulin poisoning: mendelian randomization of human cohorts and experimental validation in rat models. Int J Mol Sci. https://doi.org/10.3390/ijms26094152
8. Yu Z, Yuan Y, Zhang J, Li Y, Wang Z, Wang Y, Duan Y, Zhou Y (2024) Review of the lethal mechanism of insulin poisoning and the characteristic of forensic identification. Legal Med 70:102478. https://doi.org/10.1016/j.legalmed.2024.102478
9. Rzepczyk S, Dolinska-Kaczmarek K, Uruska A, Zaba C (2022) The Other Face of Insulin-Overdose and Its Effects, Toxics 10(3).10.3390/toxics10030123
10. Hatting M, Tavares C, Sharabi K, Rines AK, Puigserver P (2018) Insulin regulation of gluconeogenesis. Ann N Y Acad Sci 1411(1):21–35. https://doi.org/10.1111/nyas.13435
11. Rui L (2014) Energy metabolism in the liver. Compr Physiol 4(1):177–197. https://doi.org/10.1002/cphy.c130024
12. Lui DT, Woo YC, Chow WS, Lee CH, Lee AC, Leung EK, Tan KC, Lam KS, Lam JK (2019) Glycogenic hepatopathy as an unusual etiology of deranged liver function in a patient with type 1 diabetes: a case report. Med 98(17):e15296. https://doi.org/10.1097/MD.0000000000015296
13. Fujisaki N, Kosaki Y, Nojima T, Higaki T, Yamada T, Koga H, Gochi A, Naito H, Nakao A (2020) Glycogenic hepatopathy following attempted suicide by long-acting insulin overdose in patient with type 1 diabetes. JACEP Open 1(5):1097–1100. https://doi.org/10.1002/emp2.12093
14. Okamoto N, Onishi S, Onodera T, Tawara T, Okamoto H, Shimizu T, Oshiro A, Morishita Y, Nara S (2022) Insulin overdose complicated by treatment-induced acute hepatic steatosis in a nondiabetic patient. Acute Medicine & Surgery 9(1):e772. https://doi.org/10.1002/ams2.772
15. Tong F, Wu R, Huang W, Yang Y, Zhang L, Zhang B, Chen X, Tang X, Zhou Y (2017) Forensic aspects of homicides by insulin overdose. Forensic Sci Int 278:9–15. https://doi.org/10.1016/j.forsciint.2017.06.015
16. Yuan YH, Yu ZH, Zhang JX, Ma LD, Zhao SQ, Liu NG, Wu RQ, Zhang B, Liao XB, Chen X, He GL, Zhou YW (2025) Recommendation for forensic identification guidelines on insulin overdoes. Fa Yi Xue Za Zhi 41(2):168–175. https://doi.org/10.12116/j.issn.1004-5619.2024.340104
17. Adeva-Andany MM, Perez-Felpete N, Fernandez-Fernandez C, Donapetry-Garcia C, Pazos-Garcia C (2016) Liver glucose metabolism in humans. Biosci Rep. https://doi.org/10.1042/BSR20160385
18. Puigserver P, Rhee J, Donovan J, Walkey CJ, Yoon JC, Oriente F, Kitamura Y, Altomonte J, Dong H, Accili D, Spiegelman BM (2003) Insulin-regulated hepatic gluconeogenesis through FOXO1-PGC-1alpha interaction. Nature 423(6939):550–5. https://doi.org/10.1038/nature01667
19. Tzivion G, Dobson M, Ramakrishnan G (2011) FoxO transcription factors; regulation by AKT and 14-3-3 proteins. Biochim Biophys Acta 1813(11):1938–1945. https://doi.org/10.1016/j.bbamcr.2011.06.002

20. Xing YQ, Li A, Yang Y, Li XX, Zhang LN, Guo HC (2018) The regulation of FOXO1 and its role in disease progression. Life Sci 193:124–131. https://doi.org/10.1016/j.lfs.2017.11.030
21. Zhao X, Gan L, Pan H, Kan D, Majeski M, Adam SA, Unterman TG (2004) Multiple elements regulate nuclear/cytoplasmic shuttling of FOXO1: characterization of phosphorylation- and 14-3-3-dependent and -independent mechanisms. Biochem J 378(3):839–49. https://doi.org/10.1042/BJ20031450
22. Manetti AC, Visi G, Spina F, De Matteis A, Del DF, Turillazzi E, Maiese A (2022) Insulin and Oral Hypoglycemic Drug Overdose in Post-Mortem Investigations: A Literature Review, Biomedicines 10(11).10.3390/biomedicines10112823
23. Woreta TA, Alqahtani SA (2014) Evaluation of abnormal liver tests. Med Clin North Am 98(1):1–16. https://doi.org/10.1016/j.mcna.2013.09.005
24. Link W (2019) Introduction to FOXO biology. Methods Mol Biol 1890:1–9. https://doi.org/10.1007/978-1-4939-8900-3_1
25. Lee S, Dong HH (2017) FoxO integration of insulin signaling with glucose and lipid metabolism. J Endocrinol 233(2):R67–R79. https://doi.org/10.1530/JOE-17-0002
26. Radziuk J, Pye S (2001) Hepatic glucose uptake, gluconeogenesis and the regulation of glycogen synthesis. Diabetes-Metab Res 17(4):250–272. https://doi.org/10.1002/dmrr.217
27. Li X, Monks B, Ge Q, Birnbaum MJ (2007) Akt/PKB regulates hepatic metabolism by directly inhibiting PGC-1alpha transcription coactivator. Nature 447(7147):1012–6. https://doi.org/10.1038/nature05861
28. Pan CJ, Lei KJ, Annabi B, Hemrika W, Chou JY (1998) Transmembrane topology of glucose-6-phosphatase. J Biol Chem 273(11):6144–6148. https://doi.org/10.1074/jbc.273.11.6144
29. Icard P, Wu Z, Alifano M, Fournel L (2019) Gluconeogenesis of cancer cells is disrupted by citrate. Trends Cancer 5(5):265–266. https://doi.org/10.1016/j.trecan.2019.03.002
30. Dong X, Biswas A, Suel KE, Jackson LK, Martinez R, Gu H, Chook YM (2009) Structural basis for leucine-rich nuclear export signal recognition by CRM1. Nature 458(7242):1136–1141. https://doi.org/10.1038/nature07975
31. Dong X, Biswas A, Chook YM (2009) Structural basis for assembly and disassembly of the CRM1 nuclear export complex. Nat Struct Mol Biol 16(5):558–560. https://doi.org/10.1038/nsmb.1586
32. Kim DH, Perdomo G, Zhang T, Slusher S, Lee S, Phillips BE, Fan Y, Giannoukakis N, Gramignoli R, Strom S, Ringquist S, Dong HH (2011) FoxO6 integrates insulin signaling with gluconeogenesis in the liver. Diabetes 60(11):2763–2774. https://doi.org/10.2337/db11-0548
33. Fu H, Subramanian RR, Masters SC (2000) 14-3-3 proteins: structure, function, and regulation. Annu Rev Pharmacol Toxicol 40:617–47. https://doi.org/10.1146/annurev.pharmtox.40.1.617
34. Ji L, Wang Q, Liu M, Zhu C, Xiao Y, Han J, Fang Y, Ye J, Yin J, Wei L (2021) The 14-3-3 protein YWHAB inhibits glucagon-induced hepatic gluconeogenesis through interacting with the glucagon receptor and FOXO1. FEBS Lett 595(9):1275–1288. https://doi.org/10.1002/1873-3468.14063
35. Brunet A, Kanai F, Stehn J, Xu J, Sarbassova D, Frangioni JV, Dalal SN, DeCaprio JA, Greenberg ME, Yaffe MB (2002) 14-3-3 transits to the nucleus and participates in dynamic nucleocytoplasmic transport. J Cell Biol 156(5):817–28. https://doi.org/10.1083/jcb.200112059

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